

ANNUAL TUBERCULOSIS DATABASE REPORT

JANUARY - DECEMBER

2025

STRATEGIC INFORMATION COMPONENT HIV/STI/TB UNIT

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Overview

This report provides an assessment of the tuberculosis (TB) database, focusing on the quality and completeness of data captured across key domains, including patient registration, diagnostics, clinical status, medication, and demographics. A total of 53 TB cases were recorded in the database for the period January to December 2025, which is seven (7) fewer cases than reported by the National Surveillance Unit (NSU) for the same period. This discrepancy may be partially attributed to operational disruptions associated with Hurricane Melissa, which affected Jamaica during October. These disruptions contributed to delays in reporting, and timely data entry into the database, due to wide spread damage particularly in Western Jamaica, where health services and activities were temporarily impacted.

The review indicates that while certain areas, such as patient registration, clinical status, and prescription data, were relatively well-documented, significant gaps in data quality remain. For example, 15% of cases lacked a recorded medical facility at registration, treatment outcomes were documented for only 45% of patients, and co-morbidities were captured for just 21%. Diagnostic information primarily included X-rays (44 cases), smears (40 cases), and Mantoux tests (36 cases), with limited use of PCR (10 cases) and other culture-based methods (4 cases). The existing gaps in TB testing is likely impacted by the ongoing issues with the biosafety cabinet, including Heating, Ventilation, and Air-Conditioning (HVAC) deficiencies, have rendered it out of use, posing safety risks and further limiting the ability to perform essential culture-based testing.

Regarding medication, drug prescriptions were recorded for 85% of cases, whereas drug susceptibility testing was documented for only 13%, significantly constraining the capacity to detect and monitor drug-resistant TB. Demographic information, including addresses (96%), emergency contacts (94%), and social history (77%), was largely complete; however, incarceration status was inconsistently captured, creating challenges for patient follow-up and risk assessment.

Overall, the review highlights both strengths and weaknesses in the TB database. While the system effectively captures core patient and treatment information, targeted improvements are needed to enhance data completeness, accuracy, and reliability. Strengthening these areas is essential for effective patient management, robust surveillance, and program monitoring particularly in tracking treatment outcomes, identifying high-risk populations, and detecting emerging drug resistance patterns.

Tuberculosis Burden

A review of the database revealed that a total of 83 entries have been recorded since the system went live in July of 2025. These include one case from 2022, two cases confirmed in 2023, and two cases confirmed in 2024. The majority of entries (53 cases) were confirmed in 2025. During the reporting period, entries into the tuberculosis (TB) database varied, reflecting the distribution of TB burden across the country's health regions and parishes (see **Figures 1 and 2**). Among cases confirmed in 2025, the South East Regional Health Authority accounted for the highest number (39), followed by the Western and North East Regional Health Authorities, which together contributed to a total of 53 cases. At the parish level, Kingston recorded the highest number of cases (37), followed by St. James (8), St. Ann (6), and St. Catherine (2).

Figure 1. Tuberculosis cases by Region 2025

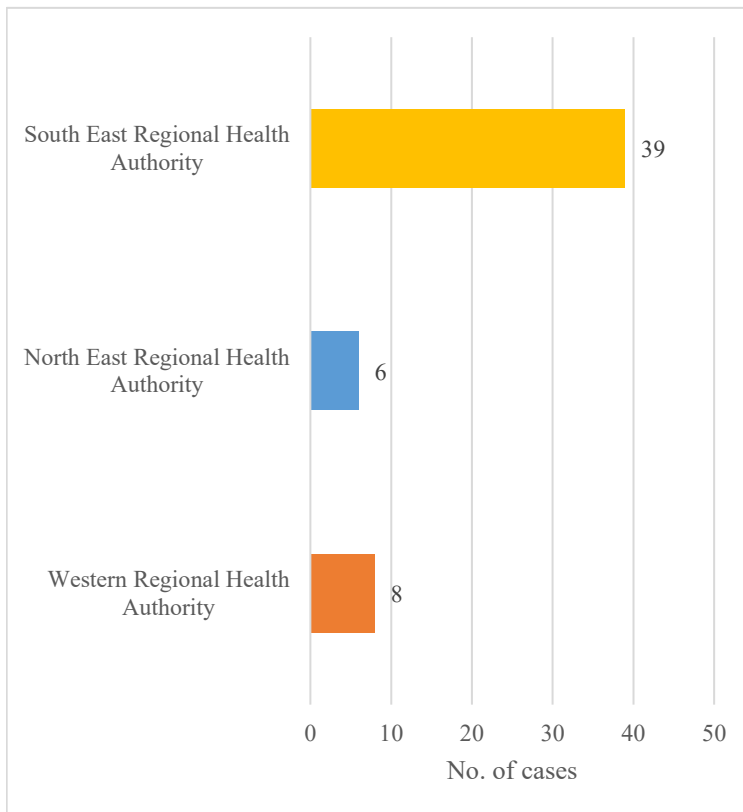
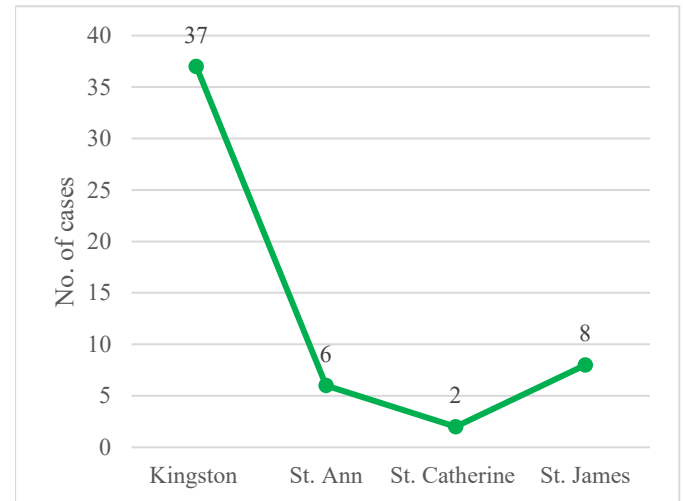


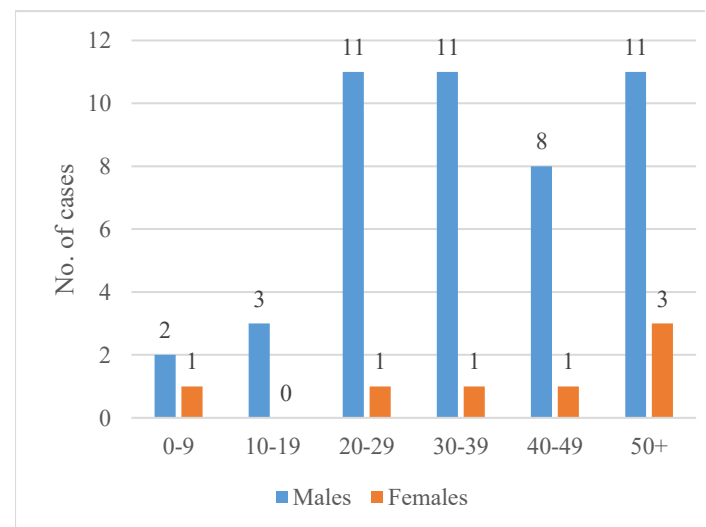
Figure 2. Tuberculosis cases by Parish 2025



Age & Sex Distribution

The age sex distribution shows that males accounted for the majority of cases across all age groups (**Figure 3**). The highest number of male cases was recorded among persons aged 20-29, 30-39, and 50 years and older (11 cases each), followed by those aged 40-49 (8 cases). Female cases were consistently lower, with the highest number observed in the 50 years and older age group (3 cases). Overall, case burden increased with age, particularly among males.

Figure 3. Age & Sex distribution of Tuberculosis cases confirmed in 2025



1. Demographics

The demographic section of the database captures patient addresses, emergency contacts, social history, and incarceration status. Of the 53 cases reviewed, addresses were recorded for 52 cases (96%), and emergency contact details were available for 50 cases (94%). Social history was documented for 41 cases (77%), while incarceration status was not recorded for any of the cases (0%) (see **Figure 4**). However, a deeper review of the social history showed that most clients lived with family (17 cases), followed by those who were incarcerated (14 cases). A notable proportion of cases had no living arrangement recorded (13 cases), highlighting gaps in data completeness. Eight cases were reported among individuals living alone, while only one case explicitly stated having no fixed living arrangement. Overall, the data suggest that most cases occurred among individuals residing with family or within institutional settings, with missing information remaining a limitation (see **Figure 4**).

Figure 4. Demographic data completeness

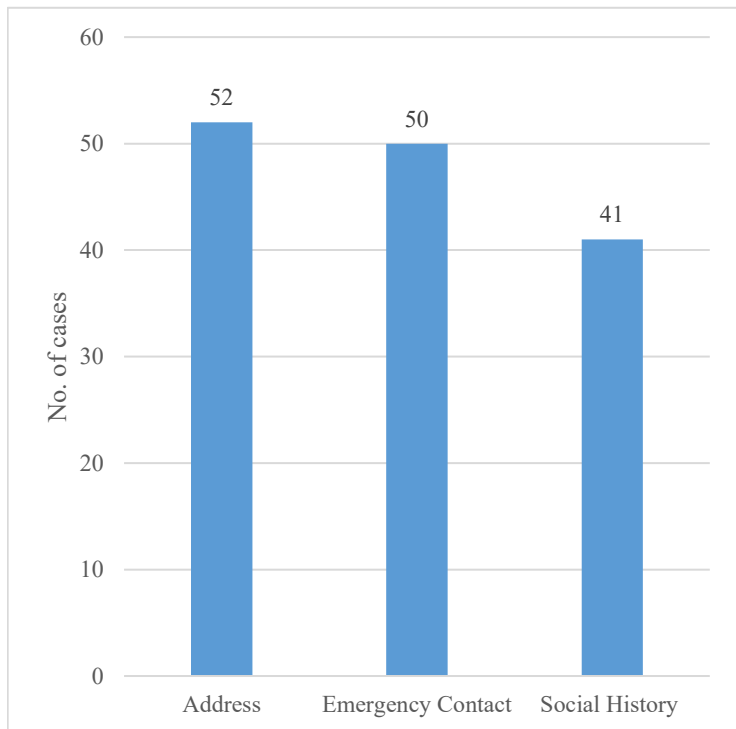
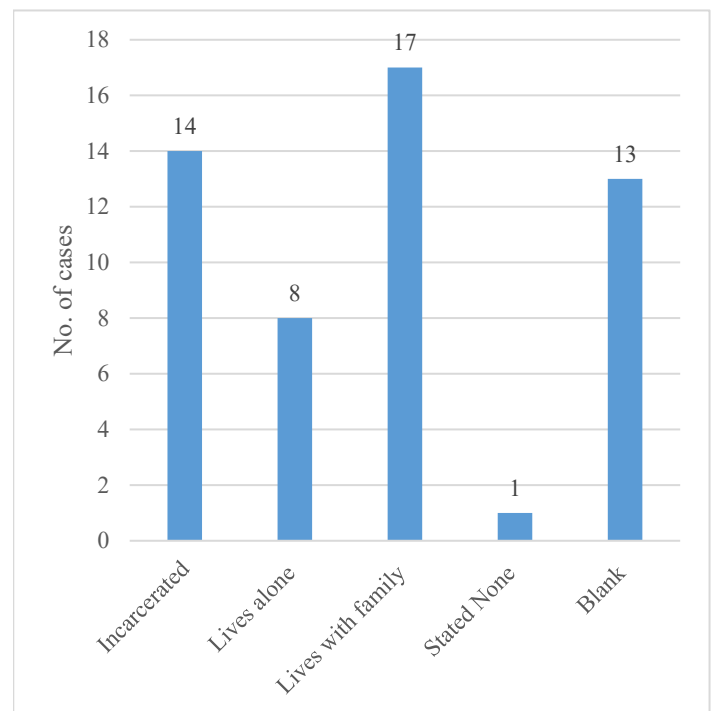
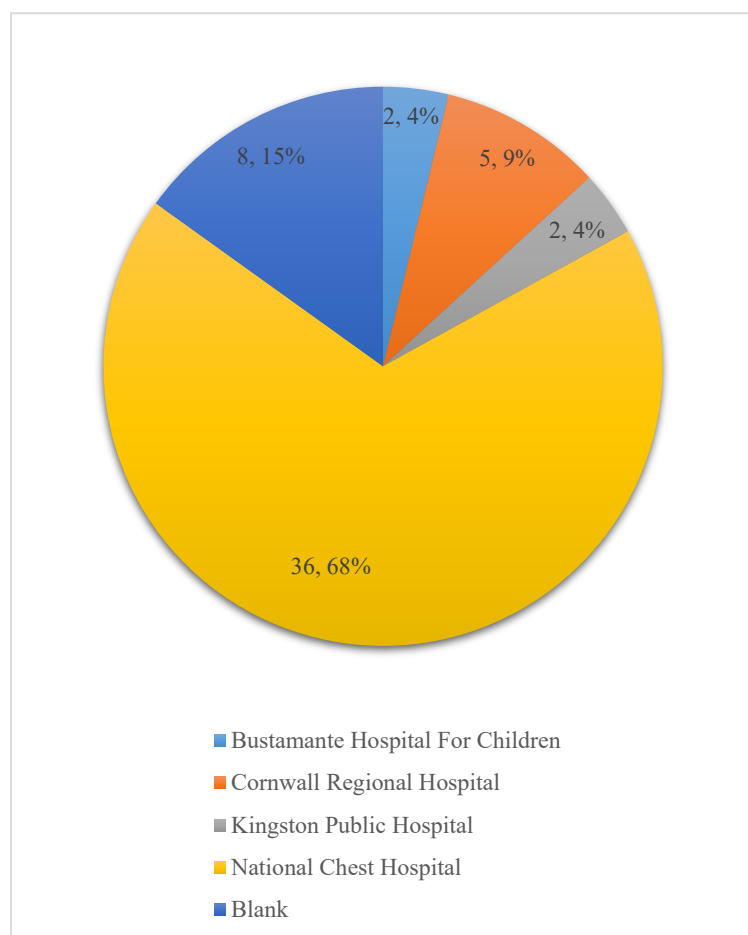


Figure 5. Living arrangement data completeness



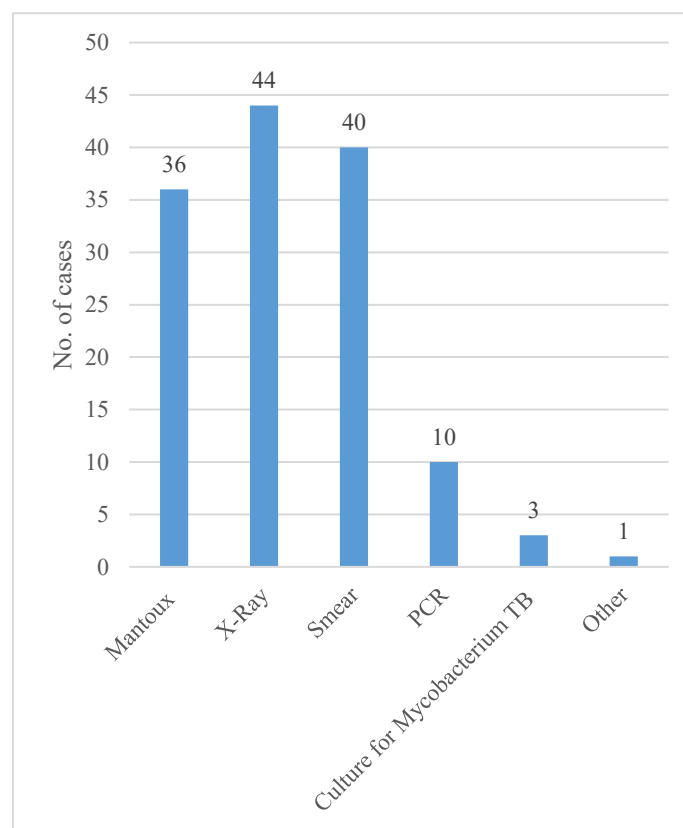
2. Registration

Registration captures the medical facility where the patient is receiving treatment. Indicators such as medical record number, appointment date, next appointment date, missed appointment, and discharge date are recorded for each patient. A review of this section showed that, among the 53 cases, most were registered at the National Chest Hospital (68%), followed by the Cornwall Regional Hospital (9%). Smaller proportions were registered at Bustamante Children's Hospital (4%) and Kingston Public Hospital (4%). However, 15% of the cases did not have a medical facility documented, indicating gaps in data completeness that may affect interpretation of facility-level burden (see **Figure 6**).

Figure 6. Medical facilities that treat TB patients

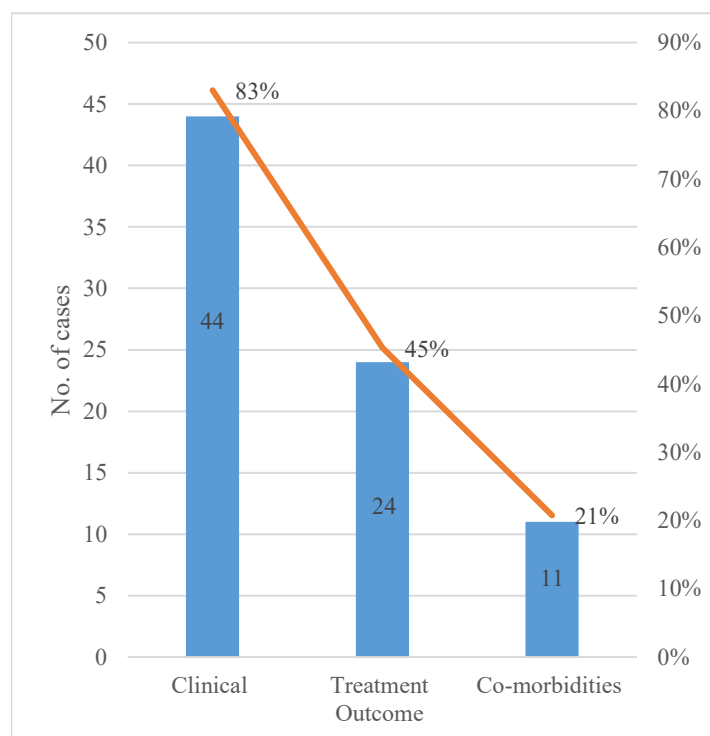
3. Investigation Results

The diagnostic methods and corresponding results used to confirm TB cases are documented in this section of the database. Several diagnostic modalities were utilized in the assessment of suspected tuberculosis cases. Chest X-ray was the most frequently performed test (44), followed by sputum smear microscopy (40) and the Mantoux test (36). Molecular testing using PCR was conducted in 10 instances, while culture for *Mycobacterium tuberculosis* was performed in 3 cases. One case indicated the use of another diagnostic method in the form of a wound swab (see **Figure 7**).

Figure 7. Types of tests

4. Patient Status

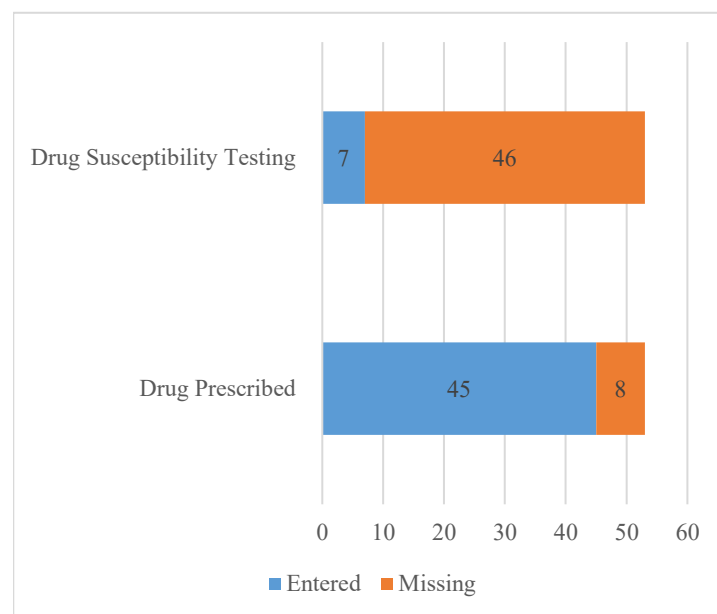
The patient status section captures information on clinical status, treatment outcomes, and the presence of co-morbidities. Documentation of clinical status was relatively high, with records completed for 44 patients (83%), suggesting consistent capture of baseline or ongoing clinical information. However, treatment outcome data were documented for 24 patients (45%), and co-morbidities were recorded for only 11 cases (21%) of which a 1 case was co-infected with HIV. The low documentation of co-morbidities hinders adequate assessment of the burden of TB-HIV co-infection and other underlying conditions (see **Figure 8**).

Figure 8. Patient status data completeness

5. Medication

This section of the database captures information on prescribed medications, dosage, and drug susceptibility testing (DST) for TB patients. Drug prescription data were available for 85% of clients, indicating a notable gap in documentation. For adult patients, the standard prescribed drug combination was Rifampicin + Isoniazid + Pyrazinamide + Ethambutol at a dosage of 600 mg + 300 mg + 1.5 g + 1.2 g, while pediatric dosages varied by age groups (600 mg + 300 mg + 1000 mg + 800 mg and 100 mg + 40 mg + 285 mg + 165 mg). Drug susceptibility testing results were recorded for only 13% of cases (see **Figure 9**). More than half of the records lacked DST data, which raises concerns from a programmatic and patient safety perspective.

More than half of the cases reviewed had no susceptibility data, which is concerning, as the absence of this information limits the ability to effectively detect and monitor drug-resistance.

Figure 9. Patient status data completeness

Recommendations

A thorough database review revealed data inconsistencies throughout all sections. Recommendations for addressing these data quality issues are provided in **Table 1** on **page 7**.

Table 1. Recommendations

Section	Challenges Identified	Recommendations	Timeline	Responsible Person
Registration	15% of cases lacked a documented medical facility	Make facility entry a mandatory field.	Ongoing	Mr. Sheldon Whorms
Demographics	Missing incarceration status; incomplete social history (e.g. living arrangements, dwelling type).	Ensure all demographic fields are mandatory; implement periodic audits to capture missing data	Ongoing	Mr. Akeem Whitehead & Mr. Andre Griffiths
Clinical Information	Incomplete treatment outcome data (33% missing); low co-morbidity documentation (especially TB-HIV).	Integrate automated alerts for overdue treatment outcome updates.	Ongoing	Mr. Sheldon Whorms
Medication & Dosage	Drug susceptibility testing documented in only 13% of cases; gaps in prescription recording	Mandate recording of DST results; to verify whether or not a test was done for the patient.	Ongoing	Mr. Sheldon Whorms
Patient Status	Treatment outcomes documented in <50% of cases; co-morbidities captured for 21%	Strengthen follow-up processes; enforce mandatory entry of treatment outcomes and co-morbidity data.	Ongoing	Mr. Akeem Whitehead & Mr. Andre Griffiths
Overall Data Quality	Overall inconsistencies across sections	Implement regular data quality checks at national level, feedback loops; consider system updates to identify inconsistencies and duplicates.	Ongoing	Strategic Information Component
		Implement data quality checks at site level	Ongoing	TB Coordinator & Medical Officers of Health
		Quarterly case reconciliation	Quarterly	National Team & National Surveillance Unit